Sai Sudharshan Ravi

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Energy Systems Engineer — Data Center Optimization — Thermal Storage

Profile

Engineer and researcher with expertise in **data center energy hubs**, **thermal energy storage**, and **energy systems optimization**. Experienced in simulation, MILP optimization, and **LCA** to design efficient, low-carbon digital infrastructure. Strong record bridging research and industry with a granted **patent**, multiple **peer-reviewed publications**, and international collaborations across Switzerland, UK, Japan, Saudi Arabia, and India.

Achievements & Highlights

- **Granted Indian Patent** (2024): Smart Remote-Controlled Fire-Fighting Hose.
- **Lead modeler** at EPFL treating **data centers as prosumers** in district energy hubs; integrated ORC/HP with **thermal storage** for net-zero strategies.
- **4+ peer-reviewed publications** on sustainable energy systems and mobility.
- Built an **urban energy hub simulation framework** for data-center integration enabling **TEA/LCA-aware** decisions.

Key Skills

- Python, R, C/C++
- Data Wrangling & LCA
- Techno-Economic Assessment
- Energy Hub Modeling, AMPL, MILP
- Communication & Leadership
- Entrepreneurship Training (UC Berkeley, Stanford)

Experience

- **EPFL (PhD Research, 2023–Present)** — Modeling **data centers as prosumers** in district energy systems; developed energy-hub optimization coupling **ORC/HP** with **thermal storage**.
- **University of Tokyo (Research Internship)** — Modeled sustainable energy pathways; applied optimization to low-carbon transport scenarios.
- **IIT Madras (Research Internship)** — Simulated clean mobility technologies; supported policy-oriented energy studies.
- **Maruti Suzuki (In-plant Training)** — Automotive systems maintenance and safety; industry-scale engineering exposure.

Selected Publications

- SS Ravi, et al. — *Modeling data centers as heat-active urban energy prosumers*, Applied Energy, 2025.
- SS Ravi, et al. — *Techno-Economic Assessment of Synthetic E-Fuels*, Energy Conversion & Management, 2023.
- SS Ravi, et al. — *EVs vs. E-Fuels: Pursuing Clean Mobility*, Science of The Total Environment, 2023.

Education

- **EPFL** — PhD in Energy Systems (2023–Present)
- **KAUST** — MS Clean Combustion, GPA 4.0/4.0 (2022–2023)
- **MIT–Anna University** — BE Automobile Engineering, GPA 9.63/10 (2017–2021)

Certifications

Python 3 Programming (Univ. of Michigan)
Data Science for Engineers (NPTEL, Elite)
Autonomous Driving Vehicles (Univ. of Toronto)

Languages

English (Full Professional) • French (Intermediate B2) • Tamil (Native)